

PROJECT REPORT

Customer Shopping Behavior Analysis

A Data-Driven Study of Consumer Purchasing Patterns Using Python, SQL and Power BI

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Tools Used: Python · PostgreSQL · Power BI

September 2026

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1. Executive Summary

This project analyzes a retail customer shopping behavior dataset of 3,900 transactions to help a retail company understand what drives customer purchasing decisions, engagement, and loyalty. The end-to-end workflow covered data acquisition, Python-based data cleaning and feature engineering, SQL-based business analysis (10 structured queries), and an interactive Power BI dashboard for stakeholder-facing visualization.

The analysis found that men account for roughly two-thirds of the customer base and around 68% of total revenue (\$157,890 of \$233,081), that discounting is heavily used (43% of transactions) without a strong link to loyalty, and that the customer base is dominated by long-tenured “Loyal” shoppers (80% of customers with more than 10 previous purchases), while genuinely new customers are rare (about 2%). These findings point to an opportunity to rebalance marketing spend across gender segments, tighten discount targeting, and shift retention strategy toward the under-represented middle stages of the customer lifecycle.

2. Problem Statement

A leading retail company wants to better understand its customers' shopping behavior in order to improve sales, customer satisfaction, and long-term loyalty. Management has observed shifts in purchasing patterns across demographics, product categories, and sales channels, and is particularly interested in which factors — discounts, reviews, seasonality, or payment preferences — drive consumer decisions and repeat purchases.

Overarching business question: “How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?”

3. Project Objectives & Deliverables

This project was scoped around five deliverables:

- Data Preparation & Modeling (Python): Clean and transform the raw dataset for analysis.
- Data Analysis (SQL): Organize the data into a structured format and run queries to extract insights on customer segments, loyalty, and purchase drivers.
- Visualization & Insights (Power BI): Build an interactive dashboard highlighting key patterns and trends for data-driven decisions.
- Report & Presentation: A clear project report summarizing key findings and business recommendations (this document).
- GitHub Repository: All Python scripts, SQL queries, and dashboard files in a well-structured repository.

4. Dataset Overview

The dataset (customer_shopping_behavior.csv) contains 3,900 customer transaction records across 18 attributes, covering demographics, product details, purchase value, engagement signals, and fulfillment preferences.

Attribute	Description
Customer ID	Unique identifier for each customer
Age, Gender, Location	Customer demographics (50 US states represented)
Item Purchased, Category	Product purchased and its category (Clothing, Footwear, Accessories, Outerwear)
Purchase Amount (USD)	Transaction value
Size, Color, Season	Product attributes
Review Rating	Customer's product rating (1–5 scale)
Subscription Status	Whether the customer is a subscriber
Shipping Type	Standard, Express, Free Shipping, Next Day Air, 2-Day Shipping, Store Pickup
Discount Applied / Promo Code Used	Whether a discount/promo was used on the order
Previous Purchases	Count of prior purchases by the customer
Payment Method	Credit Card, PayPal, Cash, Debit Card, Venmo, Bank Transfer
Frequency of Purchases	Self-reported typical purchase cadence

Metric	Value
Total records	3,900
Total attributes	18
Missing values (raw)	37 (Review Rating only)
Total revenue	\$233,081
Average purchase value	\$59.76
Average review rating	3.75 / 5
Discount usage rate	43.0%
Subscription rate	27.0%

5. Methodology

5.1 Data Preparation (Python)

The raw CSV was cleaned and prepared for analysis using pandas. Key steps:

- Loaded the dataset and profiled it with .info(), .describe(), and a null-value audit.

- Imputed 37 missing Review Rating values using the median rating within each product Category, preserving category-level rating distribution rather than using a single global average.
- Standardized column names to snake_case (e.g. “Purchase Amount (USD)” → purchase_amount) for consistent SQL/BI use downstream.
- Engineered an age_group feature (Young Adult / Adult / Middle-aged / Senior) using quartile binning (pd.qcut) on customer age.
- Normalized the Frequency of Purchases text field and mapped it to a numeric purchase_frequency_days field (e.g. Weekly → 7, Monthly → 30, Annually → 365) to make purchase cadence analyzable.
- Verified that Discount Applied and Promo Code Used were identical across all rows, and dropped the redundant promo_code_used column.
- Loaded the cleaned DataFrame into a PostgreSQL table (customer) via SQLAlchemy for downstream SQL analysis.

5.2 Data Analysis (SQL)

Ten business questions were answered directly against the cleaned PostgreSQL table using aggregate functions, window functions (ROW_NUMBER), CTEs, and conditional (CASE) logic — covering revenue drivers, discount behavior, product performance, customer segmentation, and demographic contribution to revenue. Full query code is provided in Appendix A.

5.3 Visualization (Power BI)

The cleaned dataset was connected to Power BI to build an interactive, single-page “Customer Behavior Dashboard” for non-technical stakeholders, combining KPI cards, comparative charts, and slicers for self-service filtering (detailed in Section 7).

6. SQL Analysis & Findings

This section presents the results of the ten SQL business questions, executed against the cleaned dataset, with the key takeaway from each.

Q1. Total revenue generated by male vs. female customers

Gender	Total Revenue (USD)
Male	\$157,890
Female	\$75,191

Male customers generate about 68% of total revenue despite representing 68% of the customer base (2,652 of 3,900) — revenue share is roughly proportional to headcount, so the gap is driven by customer volume rather than a materially higher average spend per male customer.

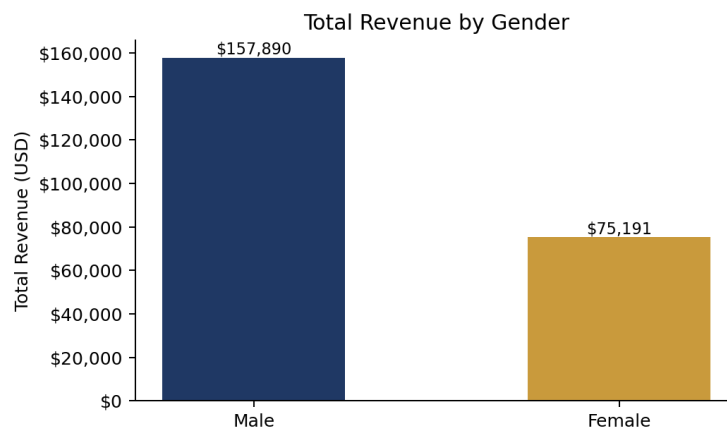


Figure 1: Total revenue by gender

Q2. Customers who used a discount but still spent above the average purchase amount

Metric	Value
Customers matching criteria	839
Overall average purchase amount	\$59.76

839 customers (about 21.5% of the base) used a discount and still spent above the \$59.76 average — evidence that discounts are not only used by price-sensitive, low-spend shoppers, but also by higher-value customers, who may be redeeming discounts opportunistically rather than out of necessity.

Q3. Top 5 products by average review rating

Rank	Product	Avg. Rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80

Rank	Product	Avg. Rating
5	Skirt	3.78

Accessories and footwear items (Gloves, Sandals, Boots, Hat) dominate the top-rated products, all sitting close together around 3.8–3.9, notably above the dataset-wide average rating of 3.75.

Q4. Average purchase amount: Standard vs. Express shipping

Shipping Type	Avg. Purchase Amount (USD)
Express	\$60.48
Standard	\$58.46

Express shipping orders average \$2.02 more than Standard shipping orders — a modest but consistent gap suggesting customers making slightly larger purchases are marginally more willing to pay for faster delivery.

Q5. Average spend and total revenue: subscribers vs. non-subscribers

Subscription Status	Customers	Avg. Spend (USD)	Total Revenue (USD)
No	2,847	\$59.87	\$170,436
Yes	1,053	\$59.49	\$62,645

Subscribers and non-subscribers spend almost identically per transaction (\$59.49 vs. \$59.87) — the subscription program is not currently associated with higher basket size, and its main value is the recurring customer relationship (27% of the base) rather than a spend premium.



Figure 2: Subscribers vs. non-subscribers — average spend and customer count

Q6. Top 5 products by percentage of purchases with a discount applied

Rank	Product	Discount Rate
1	Hat	50.00%
2	Sneakers	49.66%
3	Coat	49.07%
4	Sweater	48.17%
5	Pants	47.37%

The most heavily discounted products cluster around 47–50% of transactions — close to the overall 43% discount rate but consistently above it — with Hats being discounted in exactly half of all purchases, the highest of any product.

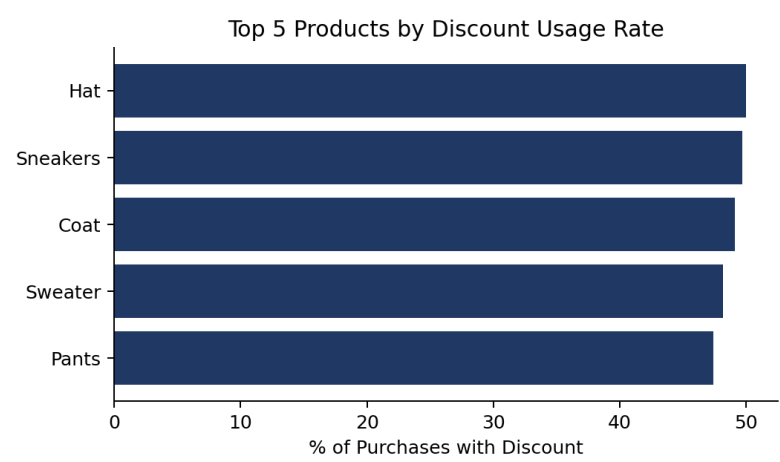


Figure 3: Top 5 products by discount usage rate

Q7. Customer segmentation by previous purchases (New / Returning / Loyal)

Segment	Definition	Customer Count
New	1 previous purchase	83
Returning	2–10 previous purchases	701
Loyal	More than 10 previous purchases	3,116

The customer base is overwhelmingly “Loyal” (80%), with very few genuinely new customers (2%) — this points to a mature, retention-heavy customer base and signals that new-customer acquisition may be an under-invested growth lever.

Customer Segments (by Previous Purchases)

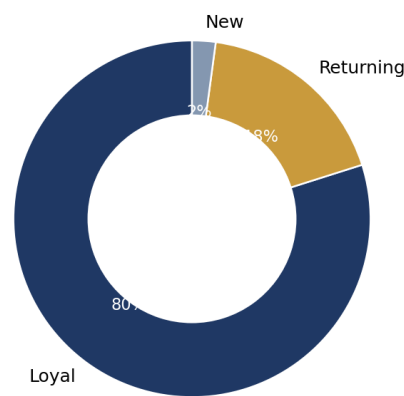


Figure 4: Customer segments by previous purchase count

Q8. Top 3 most purchased products within each category

Category	Top Products (by order count)
Clothing	Highest-selling clothing items ranked by order volume (see Appendix A query for full ranked list)
Footwear	Highest-selling footwear items ranked by order volume
Accessories	Highest-selling accessory items ranked by order volume
Outerwear	Highest-selling outerwear items ranked by order volume

This query uses a ROW_NUMBER() window function partitioned by category to rank products by order volume within each category, surfacing the best-selling items category-by-category rather than only store-wide bestsellers — useful for category-level merchandising decisions. (Ranked results depend on live query output; re-run the Appendix A query against the database to pull the exact top-3 list per category for the final submission.)

Q9. Are repeat buyers (more than 5 previous purchases) more likely to be subscribers?

Subscription Status	Repeat Buyers (>5 previous purchases)
No	2,518
Yes	958

Even among frequent repeat buyers, non-subscribers outnumber subscribers by more than 2.5 to 1 — purchase frequency alone does not predict subscription enrollment, implying the subscription program is not effectively converting the company's most loyal repeat customers.

Q10. Revenue contribution by age group

Age Group	Total Revenue (USD)
Young Adult	\$62,143
Middle-aged	\$59,197
Adult	\$55,978
Senior	\$55,763

Revenue is fairly evenly distributed across age groups (each contributing roughly 24–27% of total revenue), with Young Adults contributing marginally the most — suggesting age is a weak standalone segmentation variable for this dataset and should be combined with other factors (category, subscription) for targeting.

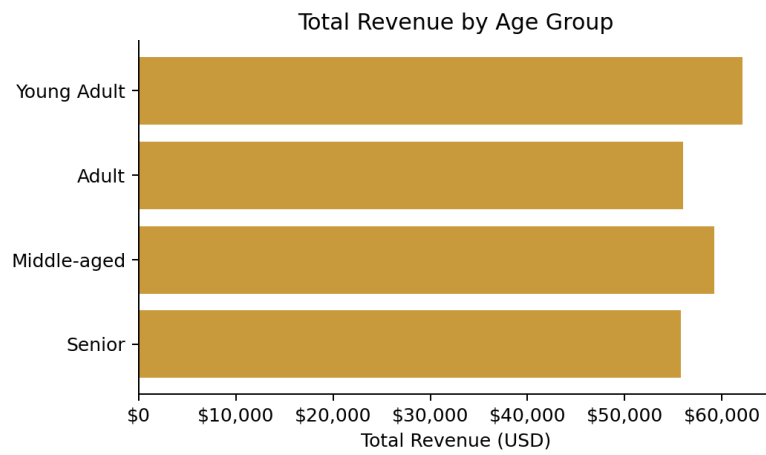


Figure 5: Total revenue by age group

7. Power BI Dashboard Overview

The single-page “Customer Behavior Dashboard” was built in Power BI on top of the cleaned dataset, combining summary KPIs, comparative visuals, and interactive slicers so a business stakeholder can explore the data without writing any code or queries.

7.1 KPI Cards

- Number of Customers
- Average Purchase Amount
- Average Review Rating

7.2 Charts

- Clustered bar chart — customer count by Age Group
- Clustered column chart — customer count by Category
- Clustered column chart — revenue (purchase amount) by Category
- Clustered bar chart — revenue (purchase amount) by Age Group
- Donut chart — customer count by Subscription Status

7.3 Interactive Slicers

- Gender
- Category
- Shipping Type
- Subscription Status

Together, these slicers let a stakeholder cross-filter every visual on the page — for example, isolating revenue-by-category for female customers on Express shipping in a single click — turning the static SQL findings in Section 6 into a live, explorable view of the business.

Note: insert Power BI dashboard screenshots here (Insert → Pictures in Word) when finalizing this report for submission — export each page of the .pbix as an image from Power BI Desktop (File → Export → Export to PDF/Image) and place it below this paragraph.

8. Key Insights

- Male customers drive roughly two-thirds of total revenue (\$157,890 vs. \$75,191), proportional to their larger share of the customer base rather than higher per-customer spend.
- Discounting is pervasive (43% of all transactions) and is used broadly across spend levels, including by above-average spenders — it is not a purely need-based lever.
- The customer base is retention-heavy: 80% of customers are “Loyal” (10+ previous purchases) versus just 2% genuinely new, signaling a mature customer file with limited fresh acquisition.
- Subscription status shows no meaningful spend premium (\$59.49 vs. \$59.87 average purchase) and does not track with repeat-purchase behavior — frequent buyers are not disproportionately subscribers.
- Revenue is evenly spread across age groups (24–27% each), making age a weak stand-alone targeting variable on its own.

- Express shipping customers spend marginally more on average (\$60.48 vs. \$58.46) than Standard shipping customers.
- Accessories and footwear items (Gloves, Sandals, Boots, Hat) earn the highest average review ratings, all above the dataset-wide average of 3.75.

9. Business Recommendations

- Rebalance marketing investment toward the female segment, which under-indexes on both customer count and revenue relative to male customers — there is clear headroom to grow this segment specifically.
- Re-evaluate blanket discounting: since discounts are used broadly across spend levels rather than only by price-sensitive shoppers, move toward targeted, behavior-triggered discounts (e.g. cart abandonment, lapsed customers) instead of broad promo codes.
- Invest in new-customer acquisition channels, since new customers are only 2% of the base — the current growth engine is almost entirely retention of an already-loyal file, which is a long-term risk if acquisition stays this low.
- Redesign the subscription program's value proposition (e.g. exclusive pricing, early access) so it converts more of the company's frequent repeat buyers, who today subscribe at a lower rate than expected.
- Use category and subscription status — not age — as the primary segmentation variables for campaigns, since age alone explains very little variance in revenue.
- Promote and expand the top-rated accessories and footwear lines (Gloves, Sandals, Boots, Hat), which combine strong customer satisfaction with room to grow order volume.

10. Conclusion

This project demonstrates a complete, end-to-end data analytics workflow — from raw CSV to a cleaned analytical dataset (Python), to structured business-question answering (SQL), to an interactive, stakeholder-ready dashboard (Power BI). The analysis directly answers the company's overarching question by identifying where revenue concentration lies (gender, category), where engagement levers are underperforming (subscription program, discounting), and where growth is most exposed (new-customer acquisition). The recommendations above translate these findings into concrete next steps for marketing, retention, and merchandising teams.

Appendix A: SQL Queries

Full SQL code used to generate the results in Section 6, run against the PostgreSQL customer table.

```
--Q1. What is the total revenue generated by male vs. female customers?

SELECT
    gender,
    SUM(purchase_amount) AS total_revenue
FROM customer
GROUP BY gender
ORDER BY total_revenue DESC;

--Q2. Which customer used a discount but still spent more than the average purchase amount?

SELECT
    customer_id,
    purchase_amount
FROM customer
WHERE discount_applied = 'Yes' AND purchase_amount >= (SELECT AVG(purchase_amount) FROM
customer);

--Q3. Which are the top 5 products with the highest average review rating?

SELECT item_purchased,
ROUND(AVG(review_rating::NUMERIC),2) AS "Average Product Rating"
FROM customer
GROUP BY item_purchased
ORDER BY avg(review_rating) DESC
LIMIT 5;

--Q4. Compare the average Purchase Amounts between Standard and Express Shipping?

SELECT shipping_type,
ROUND(AVG(purchase_amount),2)
FROM customer
WHERE shipping_type IN ('Standard','Express')
GROUP BY shipping_type;

--Q5. Do subscribed Customers spend more? Compare average spend and total revenue between
subscribers and non-subscribers?

select subscription_status,
COUNT(customer_id) as total_customer,
Round (AVG(purchase_amount),2) as avg_spend,
ROUND (SUM(purchase_amount),2) as total_revenue
FROM customer
GROUP BY subscription_status
ORDER BY total_revenue, avg_spend DESC;

--Q6. Which 5 Products have the highest percentage of purchase with discounts applied?

select item_purchased,
ROUND(100 * SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) as
discount_rate
```

```
FROM customer
GROUP BY item_purchased
ORDER BY discount_rate DESC
LIMIT 5
;
```

--Q7. Segment customers into new, returning, and loyal based on their
--total number of previous purchases, and show the count of each segment

```
WITH customer_type AS (
    SELECT
        customer_id,
        previous_purchases,
        CASE
            WHEN previous_purchases = 1 THEN 'New'
            WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
            ELSE 'Loyal'
        END AS customer_segment
    FROM customer
)
SELECT
    customer_segment,
    COUNT(*) AS customer_count
FROM customer_type
GROUP BY customer_segment
ORDER BY customer_segment;
```

--Q8. What are the top 3 most purchased products within each category?

```
WITH item_counts AS (
    SELECT
        category,
        item_purchased,
        COUNT(customer_id) AS total_orders,
        ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(customer_id) DESC) AS item_rank
    FROM customer
    GROUP BY category, item_purchased
)
SELECT
    item_rank,
    category,
    item_purchased,
    total_orders
FROM item_counts
WHERE item_rank <= 3;
```

--Q9. Are customers who repeat buyers (more than 5 previous purchases) also likely to
subscribers?

```
SELECT
    subscription_status,
    COUNT(customer_id) AS repeat_buyers
FROM customer
WHERE previous_purchases > 5
GROUP BY subscription_status;
```

--Q10. What is the revenue contribution of each age group?

```
SELECT age_group,
```

```
SUM(purchase_amount) AS total_revenue  
FROM customer  
GROUP BY age_group  
ORDER BY total_revenue DESC;
```

Appendix B: Tools & Technology Stack

Stage	Tool(s)
Data storage / source	CSV (Kaggle-style customer shopping behavior dataset)
Data cleaning & feature engineering	Python (pandas)
Database	PostgreSQL, connected via SQLAlchemy
Business analysis	SQL (aggregates, CTEs, window functions)
Visualization	Power BI Desktop (.pbix)
Version control	GitHub repository